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LESSON 1: Library management system

- Identify candidate classes (Books, Readers, Vouchers, Employees, Publishers)

+ **Book:** main entity, managing book information.

+ **DocGia:** person who borrows books.

+ **Fee:** records borrowing/returning.

+ **NhanVien:**System management (librarian, warehouse keeper...).

+ **NhaXuatBan:** book source information.

- Properties and behavior (methods) of each class

+ Sach

- o **Attributes:**maSach (PK, string), tenSach (string), tacGia (string), namXuatBan (int), maNXB (FK), **soLuongTon (int), trangThai (string:"Have available"/"On loan").**

- o **Behavior:**themSach (), capNhatSach (), xoaSach (), timKiemSach (), capNhatTonKho ().

+ DocGia

- o **Attributes:**maDocGia (PK, string), hoTen (string), ngaySinh (date), diaChi (string), soDienThoai (string), ngayDangKy (date), gioiTinh (boolean).

- o **Behavior:**dangKyThe (), capNhatThongTin (), xemLichSuMuon (). + PhieuMuon

- o **Attributes:**maPhieu (PK, string), maDocGia (FK), maSach (FK), maNhanVien (FK), ngayMuon (date), ngayHenTra (date), ngayTraThucTe (date), tinhTrang (string:"On loan"/"Paid"/"Overdue").

- o **Behavior:**createPhieuMuon(), search(), saveTraQuaHan(), crystalPhiPhat().

+ NhanVien

- o **Attributes:**maNhanVien (PK, string), hoTen (string), chucVu (string), ngaySinh (date), soDienThoai (string).

- o **Behavior:**dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().

+ NhaXuatBan

- o **Attributes:**maNXB (PK, string), tenNXB (string), diaChi (string), email (string).

- o **Behavior:**themNXB (), capNhatNXB ().

- Relationships

+Books and Publishers: 1 book – 1
Publishing House

- +**Readers and Borrowing Slips:** 1 reader – many borrowing slips
- +**Books and borrowing slips:** 1 book – many borrowing slips
- +**Employees and loan vouchers:** 1 employee – many loan vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- + DangKyTaiKhan (Customer registration).
- + DangNhap (Sign in).
- + SeeSanPham (Search / View product list).
- + ThemVaoGioHang (Add to cart).
- + DatHang (Order).
- + ThanhToan (Online payment / COD).
- + View DonHang (View order history).

Admin / NguoiQuanLy:

- + QuanLySanPham (Add/Edit/Delete products).
- + QuanLyDonHang (View/Confirm/Cancel order).
- + QuanLyKhachHang (Customer information management).
- + QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- + ThanhToan includes XacNhanDonHang.
- ++DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- + **Customers (Customer):** from register, view orders.
- +**SanPham (Product):** product management.
- +**GioHang (Cart):** temporarily save the product before ordering.
- + **Don Hang (Order):** from order, payment.
- +**ChiTietDonHang (OrderDetail):** product details in the order.
- +**Payment (Payment):** payment processing (may have subclasses such as OnlinePayment, COD).
- ++Admin or NguoiQuanLy (inherits from User if there is a user system).

- Analyze relationships between classes (specify inheritance if any)
- +**KhachHang and DonHang:** 1 customer – many orders
- +**DonHang and ChiTietDonHang:** 1 order – many order details (Composition)
- +**ChiTietDonHang and SanPham:** many single details – 1 product
- +**DonHang and ThanhToan:** 1 order – 1 payment
- +User (abstract class) ←inherit ←KhachHang and Admin (KhachHang and Admin are both a type of User, sharing the same account, password, full name).

Lesson 3: Convert class diagrams into data tables

1.MonHoc Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	